# Shear

Shear slides a channel to open or close space without full recompaction

# The idea

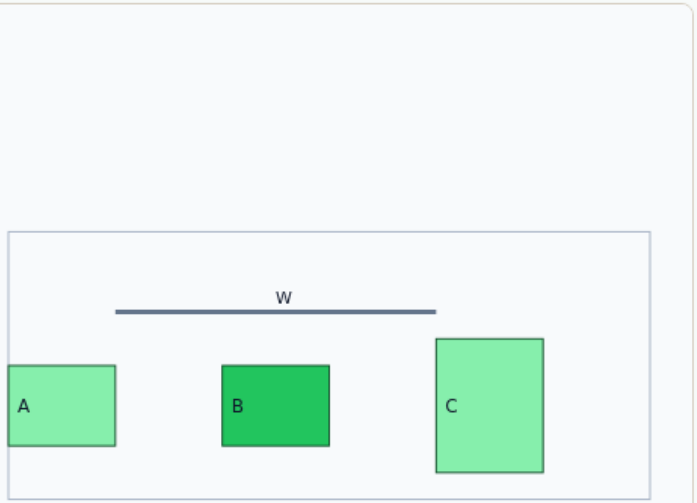
- Shear_channel_lite(x_lo,x_hi,delta_y) moves objects intersecting the band

# Channel

SHEAR

## Channel

Step 1 / 5 · ch · module03-03-shear-lite




**Explanation**

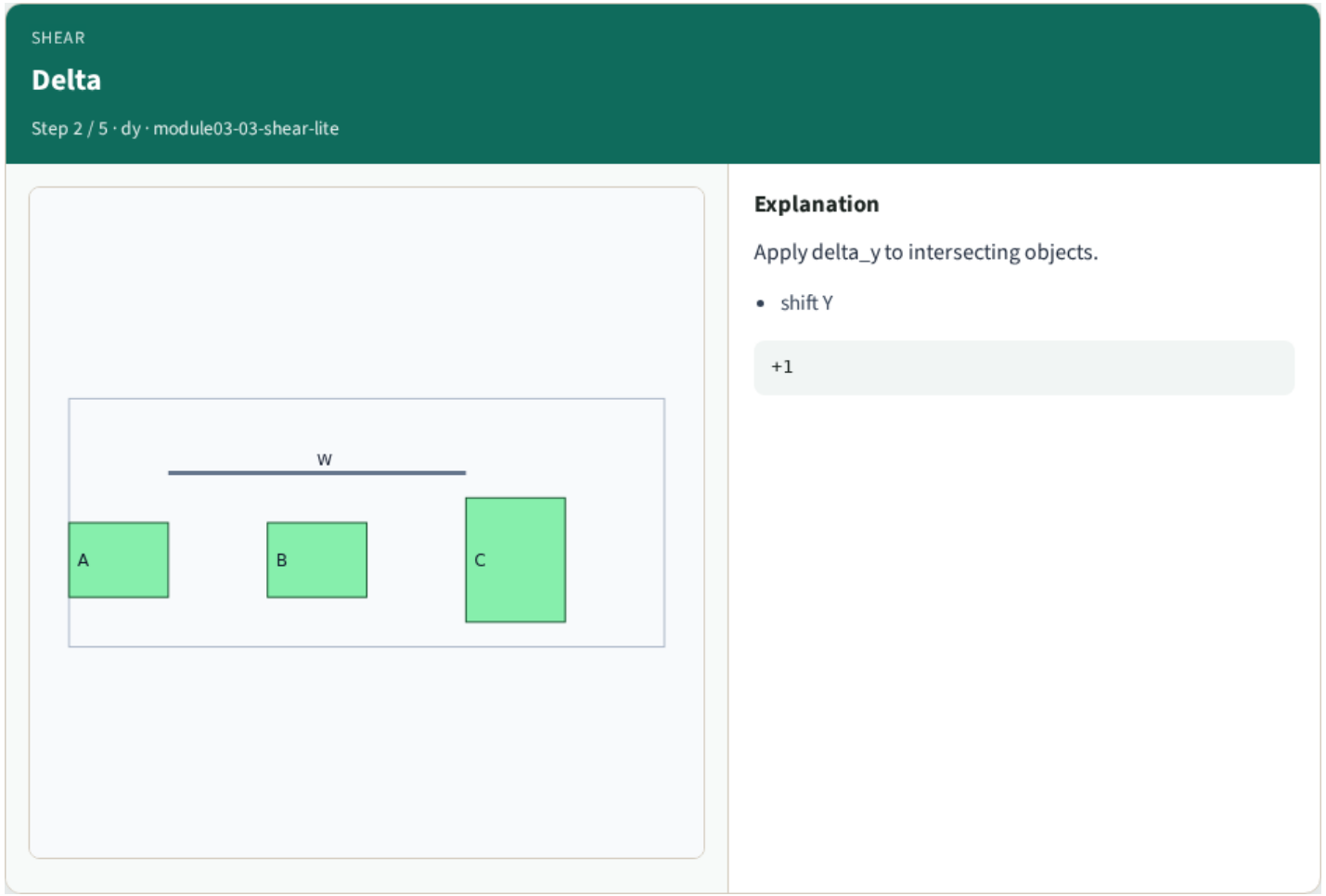
Pick an X band covering B.

- band

```
7..13
```

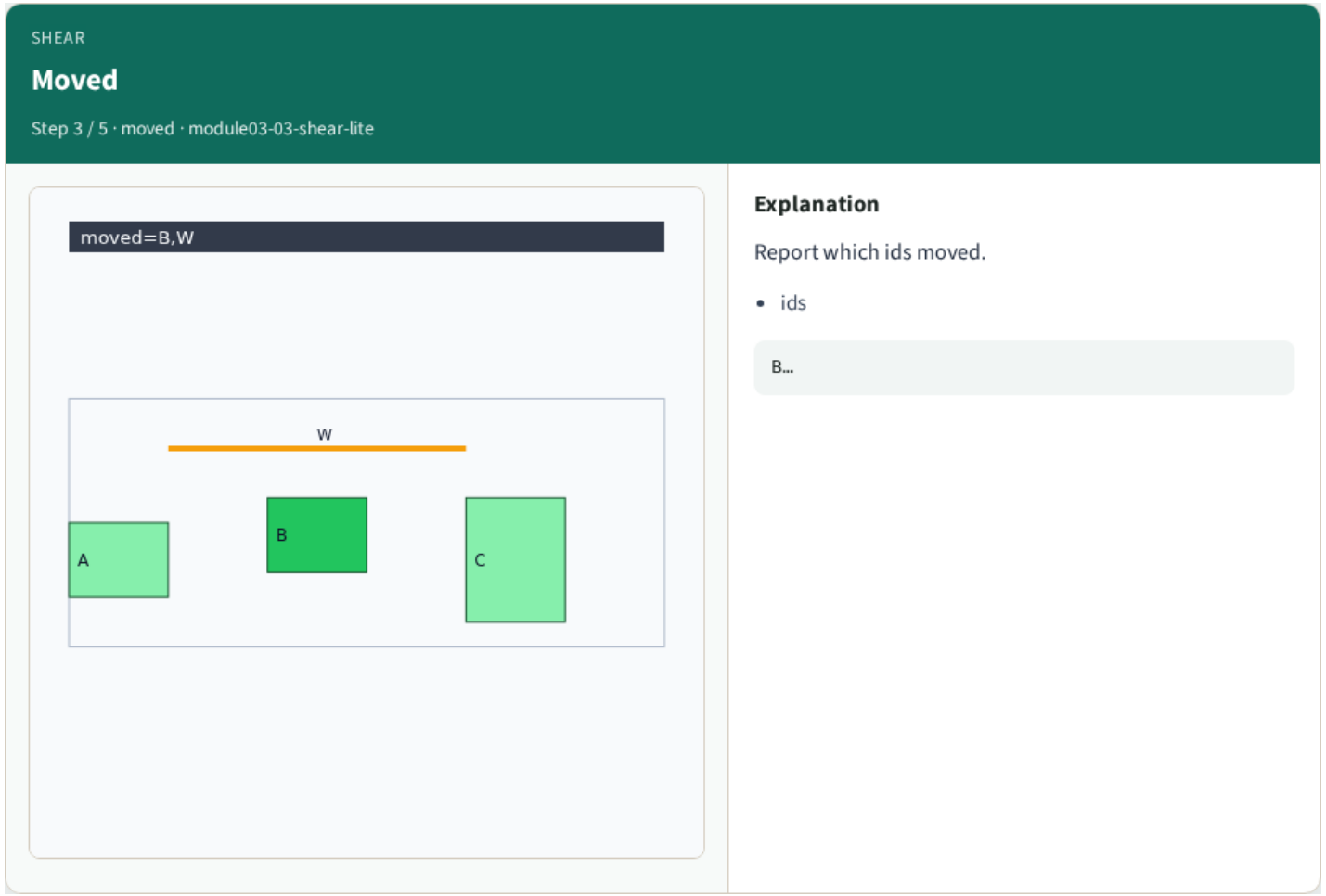
Channel

# Delta


SHEAR
Delta
Step 2 / 5 · dy · module03-03-shear-lite
W
A
B
C
Explanation
Apply delta_y to intersecting objects.
shift Y
+1

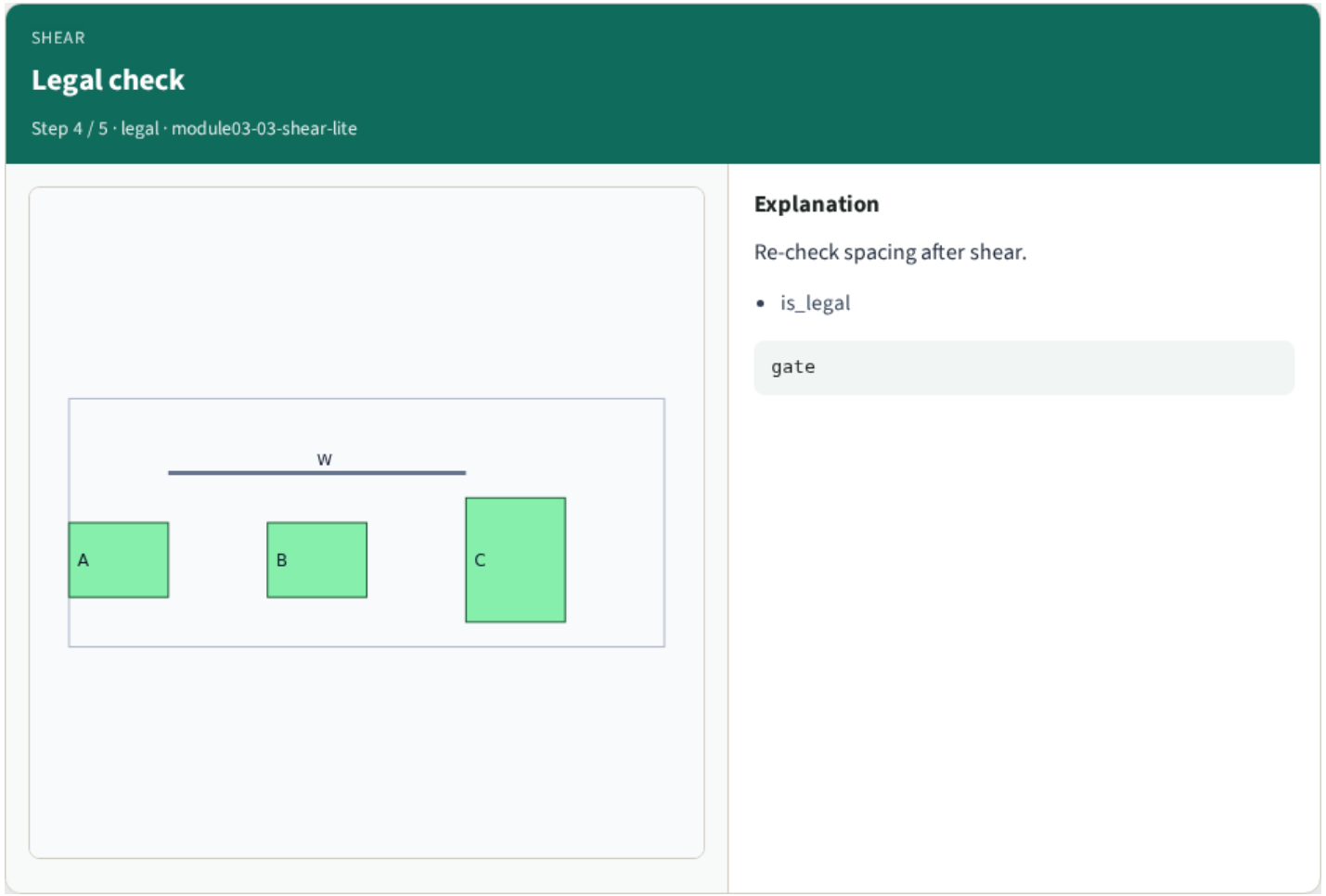
Delta

# Moved


SHEAR
Moved
Step 3 / 5 · moved · module03-03-shear-lite
moved=B,W
W
A
B
C
Explanation
Report which ids moved.
ids
B...

Moved

# Legal check



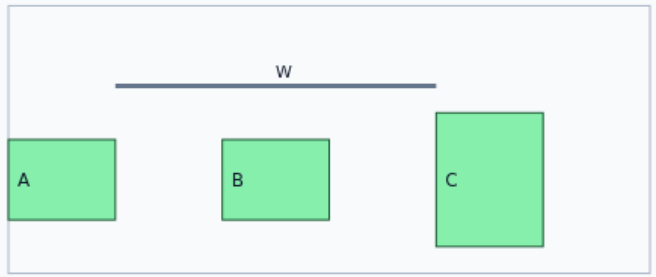

Legal check

# Use

SHEAR

## Use

Step 5 / 5 · use · module03-03-shear-lite




**Explanation**

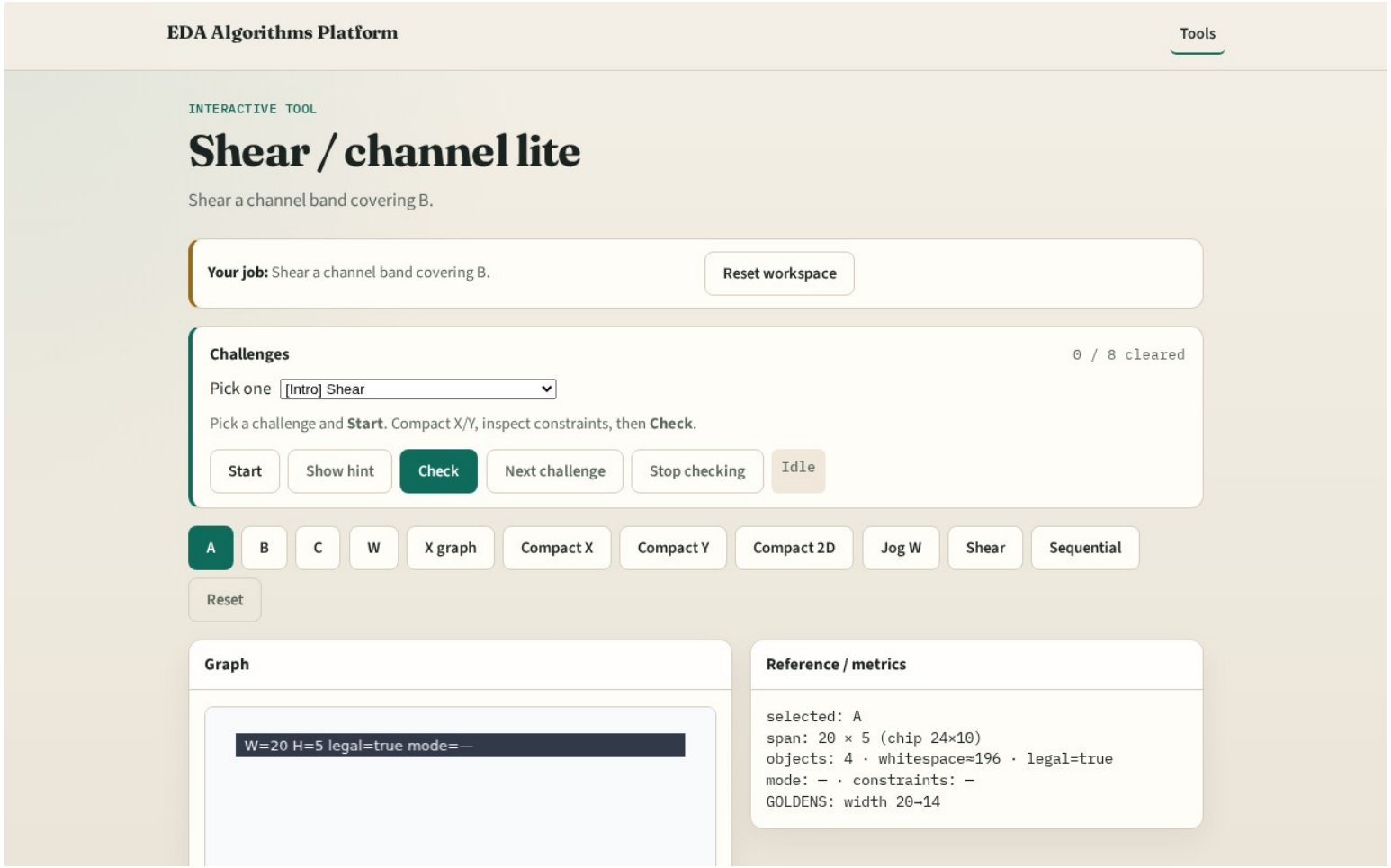
Shear is a local cleanup before another compact pass.

- iterate

```
next seq
```

Use

# Browser lab track




Browser lab starter

# Implement track

- Call shear_channel_lite(DATA, 7, 13, 1); print moved

# Pitfalls

- Shearing the whole chip blindly
- Skipping legality after shear

# Your turn

- Next: sequential compact pipeline